3NF

BCNF vs. 3NF

BCNF

Relation R is in BCNF

iff

for every FD $A \rightarrow B$ in R,

- $A \rightarrow B$ is trivial, or
- A is a superkey for R.

3NF

Relation R is in 3NF

iff

for every FD $A \rightarrow B$ in R,

- $A \rightarrow B$ is trivial, or
- A is a superkey for R, or
- every element of B is part of some key.

3NF decomposition is lossless-join

Doesn’t decompose any relation that BCNF doesn’t.

Thus has same property that

if ABC decomposes to AB and AC,

then $A \rightarrow B$ or $A \rightarrow C$.

3NF doesn't lose FD information

R(name, company, category)
{category, company} → {name}
{name} → {company}

BCNF: "Bad"
3NF: "Good" because **company** part of some key.

Keys: {category, company}, {category, name}

3NF allows some redundancies/anomalies

name	company	category
Gizmo	GizmoWorks	Gadget
Gizmo	GizmoWorks	Camera
GizmoPlus	GizmoWorks	Gadget
GizmoPlus	GizmoWorks	Camera
NewThing	NULL	Gadget

Redundancy. Repeating product's company.

Insertion anomaly. Product not yet made by any company.

BCNF vs. 3NF

BCNF

- Lossless-join
- Can lose FD info

3NF

- Lossless-join
- Doesn’t lose FD info
- Allows more redundancy
- Decomposition algorithm more complicated

Intuition for BCNF & 3NF

(3NF) “[Every] non-key field must provide a fact about the key, the whole key, and nothing but the key.”

– Bill Kent

(BCNF) “Each attribute must represent a fact about the key, the whole key, and nothing but the key.”

– Chris Date